Permit No. MI0020257

FACT SHEET

PERMITTEE/FACILITY NAME: City of Cadillac / Cadillac WWTP

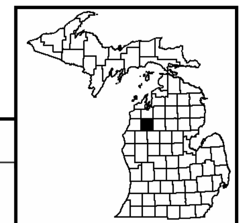
COUNTY: WEXFORD

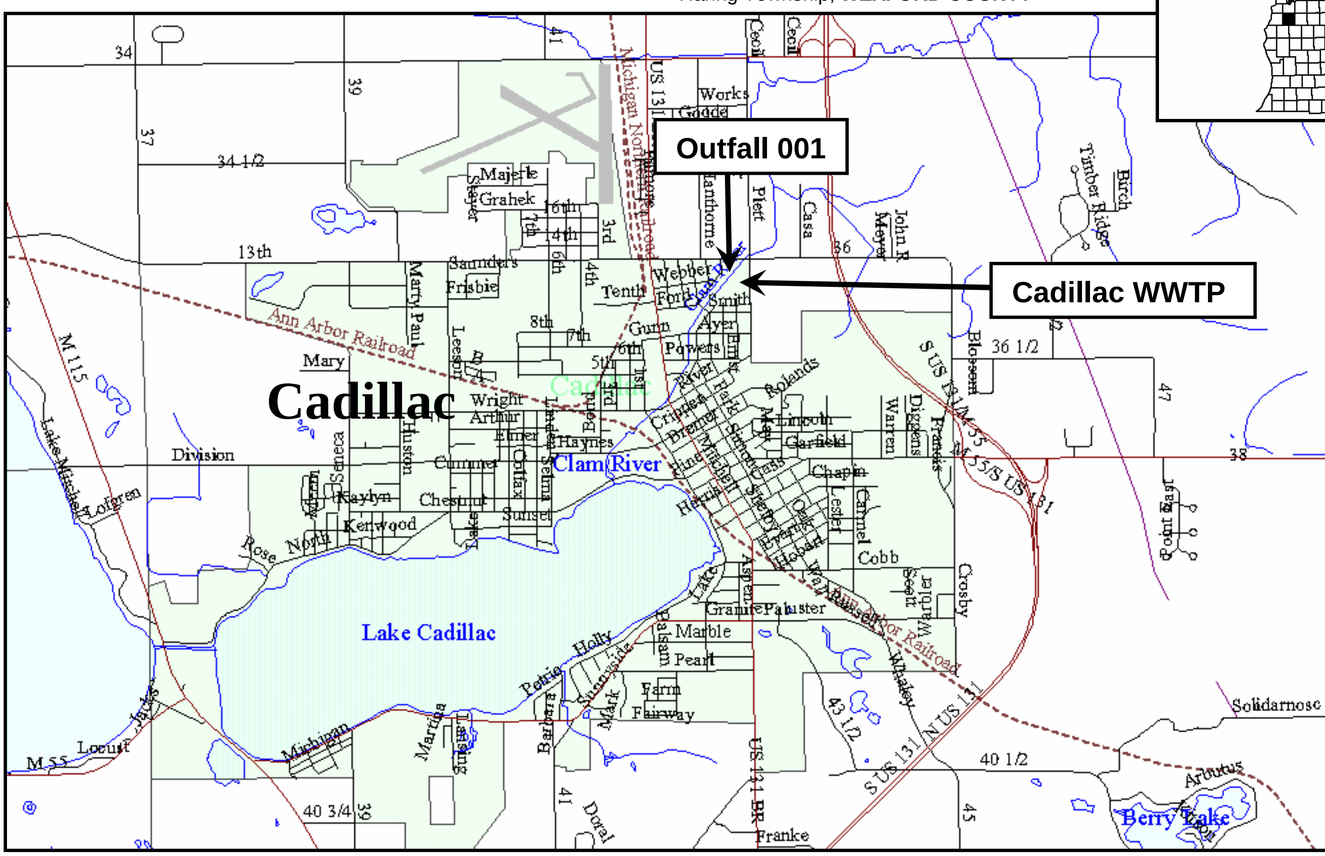
DESCRIPTION OF EXISTING WASTEWATER TREATMENT FACILITIES

The Cadillac WWTP was originally constructed in 1963 as a secondary treatment activated sludge facility. Improvements in 1975, 1990, and 1996 added Rotating Biological Discs (RBCs) and fine bubble diffusers to improve biological treatment and ammonia removal; additional secondary clarifiers and sand filters for solids removal; and further improvements in equalization, grit and solids handling, and biosolids storage. In 2003 the sodium hypochlorite disinfection system was replaced with an ultra violet (UV) light disinfection system. Phosphorus is removed with ferric chloride. Sludge generated at the facility is stabilized in an anaerobic digester and stored until land applied. Treated effluent and storm water collected from the facility grounds are discharged to the Clam River.

MAP OF DISCHARGE LOCATION

Facility Location Coordinates:
NE1/4, NE1/4, Section 33, Town 22N, Range 9 W
Haring Township, **WEXFORD COUNTY**





RECEIVING WATER

The Clam River is protected for agricultural uses, navigation, industrial water supply, public water supply in areas with designated public water supply intakes, warm-water fish (May – October), cold-water fish (November – April), other indigenous aquatic life and wildlife, partial body contact recreation, total body contact recreation (May through October), and fish consumption.

The receiving stream flows used to develop effluent limitations are a 95 percent exceedance flow of 2.7 cfs, a harmonic mean flow of 14 cfs, and a 90-day, 10-year low flow of 3.8 cfs.

MIXING ZONE

For toxic pollutants, the volume of the Clam River used in assuring that effluent limitations are sufficiently stringent to meet Water Quality Standards is 25 percent of the applicable design flows of the receiving stream.

For other pollutants, the volume of Clam River used in assuring that effluent limitations are sufficiently stringent to meet Water Quality Standards is the applicable design flows of the receiving stream.

EXISTING EFFLUENT QUALITY: (from DMR data from April 2004 to April 2007)

Parameter	Minimum Daily	Maximum Monthly	Maximum 7-day	Maximum Daily	Units
Flow	---	2.572	---	3.155	MGD
Carbonaceous Biochemical Oxygen Demand ($CBOD_5$)					
May 1 – Nov. 30	---	4.5	---	8.6	mg/l
Dec. 1 – Mar. 31	---	7.8	---	31	mg/l
Apr. 1 – Apr. 30	---	5.8	---	12.7	mg/l
Total Suspended Solids					
May 1 – Nov. 30	---	4.2	4.5	---	mg/l
Dec. 1 – Apr. 30	---	11.1	11.9	---	mg/l
Ammonia Nitrogen (as N)					
May 1 – Sept. 30	---	0.5	---	2.5	mg/l
Oct. 1 – Nov. 30	---	0.7	---	4.5	mg/l
Dec. 1 – Mar. 31	---	3.5	---	---	mg/l
Apr. 1 – Apr. 30	---	1.1	---	---	mg/l
Total Phosphorus (as P)					
May 1 – Mar. 31	---	0.4	---	---	mg/l
Apr. 1 – Apr. 30	---	0.46	---	---	mg/l
Fecal Coliform Bacteria	---	171	445	---	cts/100ml
Lindane	---	0.01	---	---	μg/l
Total Mercury	---	2.9	---	---	ng/l
Acute Toxicity	---	---	---	2.1	TU_A
Chronic Toxicity	---	2.1	---	---	TU_C
Total Cadmium	---	0.345	---	0.54	μg/l
Total Copper	---	11	---	14	μg/l
Total Zinc	---	363	---	929	μg/l
pH	6.1	---	---	8.1	S.U.
Dissolved Oxygen					
May 1 – Nov. 30	3.2	---	---	---	mg/l
Dec. 1 – Apr. 30	6.0	---	---	---	mg/l

PROPOSED EFFLUENT LIMITATIONS: (see draft permit)

BASIS FOR PROPOSED EFFLUENT LIMITATIONS

Based on this facility's application for an NPDES discharge permit, the Michigan Department of Environmental Quality proposes to issue the applicant a permit to discharge, subject to effluent limitations and certain other conditions within the permit. Effluent limitations for $CBOD_5$ (Dec – Mar) and Ammonia Nitrogen (Dec – Apr) are based on Antibacksliding Regulations. Effluent limitations for $CBOD_5$ (May – Nov, Apr), Ammonia Nitrogen (May – Sept), Total Phosphorus (all year), Chronic Toxicity, and Dissolved Oxygen are Water Quality Based Effluent Limitations. Effluent limitations for Fecal Coliform Bacteria and pH are Water Quality Standards. Monitoring requirements for Total Selenium, Total Zinc, and Available Cyanide are based on Water Quality Concerns. Effluent limitations for Total Suspended Solids (May – Nov) are based on Advanced Wastewater Treatment. Effluent limitations for Total Suspended Solids (Dec – Apr) and Total Suspended Solids Minimum Percent Removal (Dec – Apr) are Secondary Treatment Standards. Monitoring requirements for Flow and Total Mercury are based on Permit Writer's Judgment. Effluent limitation for Total Mercury is a Water Quality Variance.

REGISTER OF INTERESTED PERSONS

Any person interested in a particular application or group of applications, may leave his/her name, address, and telephone number as part of the file for an application. The list of names will be maintained as a means for persons with an interest in an application to contact others with similar interests.

PUBLIC COMMENT

Comments or objections to the draft permit received between ______________________
and ______________________ will be considered in the final decision to issue the permit.

If submitted comments indicate significant public interest in the application or if useful information may be produced, the Michigan Department of Environmental Quality, at its discretion, may hold a public hearing on the application. Any person may request the Michigan Department of Environmental Quality to hold a public hearing on the application. The request should include specific reasons for the request, indicating which portions of the application or draft permit constitutes the need for a hearing.

Public notice of a hearing will be provided at least thirty (30) days in advance. The hearing will normally be held in the vicinity of the discharge. The Michigan Department of Environmental Quality will consider comments made at the hearing when making its final determinations on the permit. Further information regarding the draft permit, and procedures for commenting or requesting a public hearing may be obtained by contacting Mr. Jeff Fischer, Permits Section, Water Bureau, Department of Environmental Quality, P.O. Box 30273, Lansing, Michigan, 48909, telephone: 517-335-4188
e-mail: fischerj1@michigan.gov.